

ROBOTIC ARM CONTROLLED BY ROBOTIC GLOVE

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Abstract

A Robotic Arm controlled by a Robotic Glove using Flex Sensor is a promising development in the field of robotics that offers a natural and intuitive way for humans to control complex machinery. This system consists of a glove worn by the human operator and a robotic arm that mimics the movement of the operator's hand. The glove is equipped with flex sensors that detect the movement of the fingers and hand and transmit this data to the robotic arm. The robotic arm then replicates the motion of the operator's hand in real-time, allowing for precise control of the machinery. One of the key advantages of this system is its ease of use. The flex sensors on the glove provide a reliable and cost-effective way to capture the motion of the operator's hand, making it an attractive option for a wide range of industries. Additionally, the system is designed to be intuitive, allowing operators to control complex machinery with minimal training. This is particularly important in industries such as manufacturing and medical procedures, where operators must be able to perform intricate tasks with high precision. Another advantage of the Robotic Arm controlled by a Robotic Glove using Flex Sensor is its potential for use in remote control of machinery. The system can be used to control machinery from a distance, allowing operators to perform tasks in hazardous or hard-to-reach locations. This could be particularly useful in industries such as mining, oil and gas, and construction, where workers often operate in dangerous environments.

Keywords: Arduino uno, Arduino Nano, Flex Sensor, Accelerometer, Bluetooth HC-05 module.

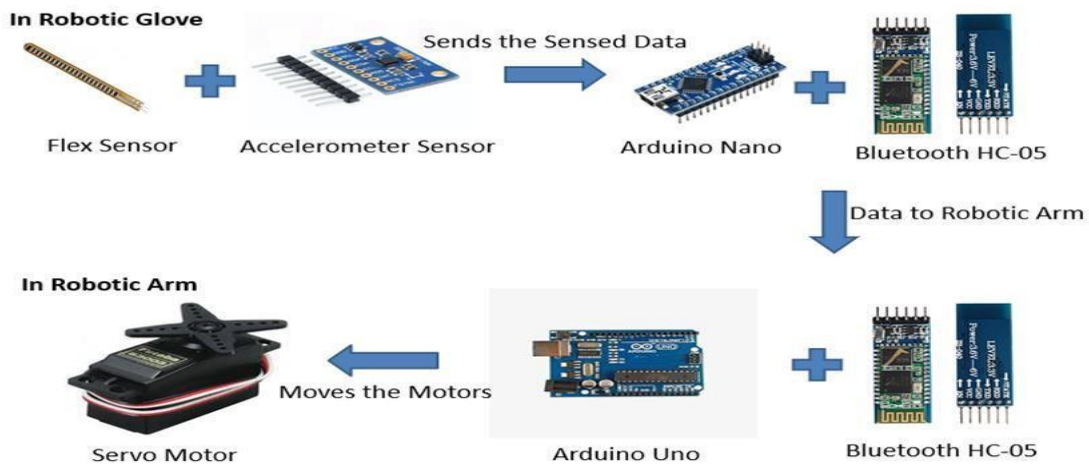
I. INTRODUCTION

Robotic Arm controlled by a Robotic Glove is a recent development in the field of robotics that offers a promising solution for natural and intuitive control of complex machinery. This technology is particularly important in industries where precision and accuracy are essential, such as manufacturing, medical procedures, and remote control of machinery. By using flex sensors embedded in a robotic glove, this system allows operators to control robotic arms with a high degree of precision and accuracy, mimicking the natural movement of the human hand. The use of flex sensors is a key innovation that has made this technology possible. These sensors are able to detect the movement of the operator's fingers and hand, allowing for real-time feedback and control of the robotic arm. This approach is much more natural and intuitive than other methods of control, such as joysticks or touchscreens, which can be cumbersome and difficult to use for precise movements. The development of a Robotic Arm controlled by a Robotic Glove has the potential to revolutionize the way we interact with machines. By providing an intuitive and natural interface for controlling complex machinery, this technology has the potential to increase efficiency, reduce errors, and improve safety in a wide range of industries. It can also enable remote control of machinery, allowing operators to perform tasks in hazardous or hard-to-reach locations without putting themselves at risk.

II. WORKING OF PROJECT

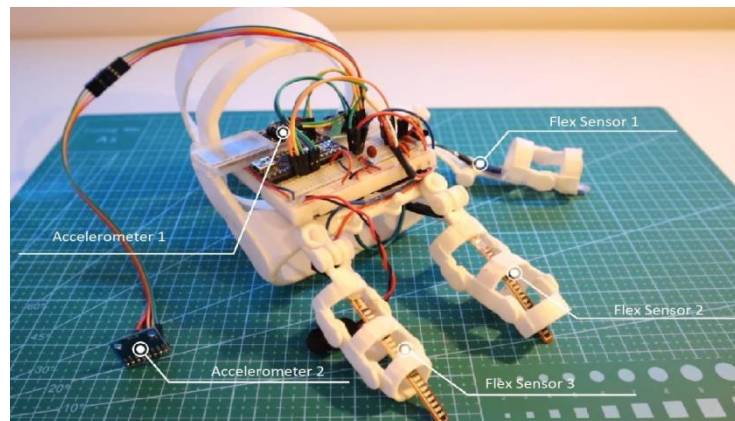
In this project Flex Sensors and the accelerometer will collect the data from the human gestures and sends then to the Arduino nano. This data will be transmitted through the Bluetooth module to another Bluetooth module which is present at the robotic arm side. The Bluetooth module sends the data to the Arduino uno. This Arduino uno further

sends this information to the servo motor drive. Here it operates the required servo motor with the required angle. Then the robotic arm starts moving according to the operator's movements.

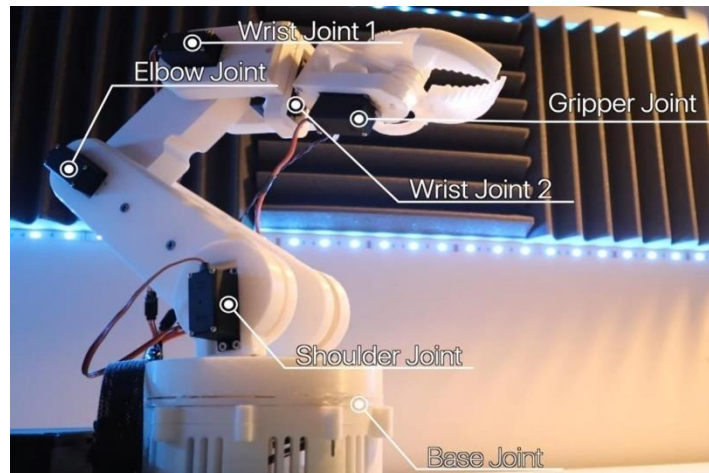


III. HARDWARE DEVICE

The robotic glove shown in the below figure consists of 3 Flex sensors and 2 Accelerometers. These Sensors are used for different kinds of functions like for Flex Sensor 1 Rotate Left/Right, Flex Sensor 2 Gripper Open/Close, Flex Sensor 3 Elbow Joint Bend/Stretch, Accelerometer 1 Rotate Clockwise and Accelerometer 2 Wristjoint2Up/Down.



The robotic arm shown in the below Figure consists of total 6 joints named as shoulder joint, base joint, elbow joint, wrist joint 1, wrist joint 2 and gripper joint. These joints work according to the gestures given by the operator by using the robotic glove.



IV. SOFTWARE ENVIRONMENT

The Arduino uno and the Arduino nano boards are programmed by using the Arduino IDE, it is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, MacOS X, and Linux. It supports the programming languages C and C++. The program or code written in the Arduino IDE is often called as sketching. We need to connect the Genuine and Arduino board with the IDE to upload the sketch written in the Arduino IDE software. The sketch is saved with the extension '.ino.'

V. FLEX SENSOR BASED ROBOTIC ARM USING MICRO CONTROLLER

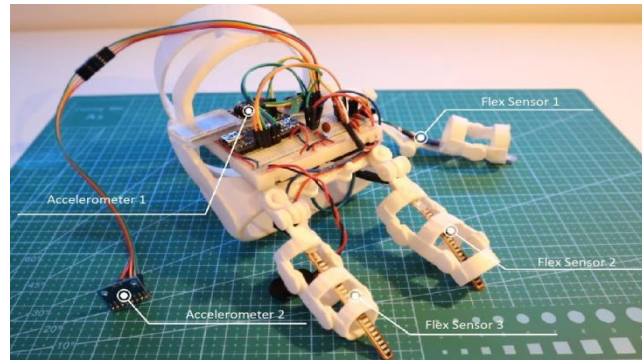
Flex sensors are analog resistors. These resistors work as variable analog voltage divider. Inside the flex sensor are carbon resistive elements with thin flexible substrate. More carbon means less resistance. When the substrate is bent the sensor produces resistance output relative to the bend radius. The flex sensor achieves great form-factor on a thin flexible substrate. When the substrate is bent, the sensor produces a resistance output correlated to the bend radius. Smaller the radius, higher will be the resistance value. The impedance buffer in the circuit is a single sided operational amplifier used with these sensors. Since low bias current of the op amp reduces error due to source impedance of the flex sensor as voltage divider. The variation in deflection or bending of flex sensor results in variation of resistance itself. The signal conditioning circuit is used to read these resistance changes and it is given to ADC.

VI. RESULTS AND OUTPUT

The project should result in a functional prototype of a robot arm that can be controlled by human gestures. This will demonstrate the feasibility of the concept and provide a basis for further development also known as **Functional prototype**. The accuracy of the gesture recognition system will be an important outcome of the project. The system should be able to recognize a range of hand gestures reliably and accurately, allowing the user to control the robot arm with ease called as **Gesture recognition accuracy**. The robot arm should be able to move smoothly and precisely in response to the user's gestures. The movement should be fluid and natural, allowing the user to control the arm with a high degree of precision is called **Smooth and precise movements**.

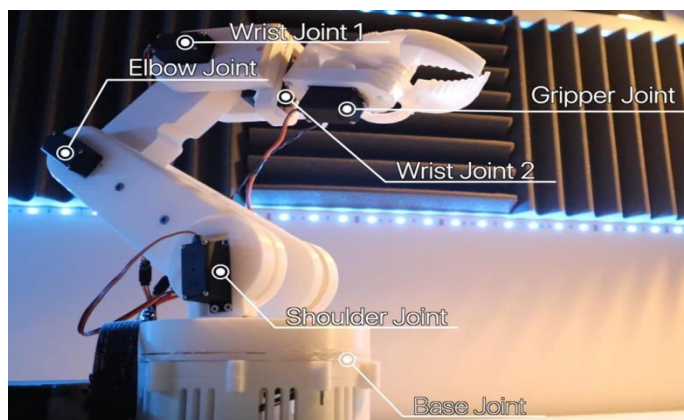
Robotic Glove

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Robotic Arm

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Controlling Robotic arm by using Robotic Glove

The below Figure is an example of how to use the robotic arm to do the small works like lifting. Here the operator used the flex sensor 2 for holding the object with the gripper of the robotic arm. By making different types of gestures using robotic glove the robotic arm reacts according to the gestures came from the robotic glove.



VII. FUTURE ENHANCEMENT

The Robotic Arm controlled by a Robotic Glove using Flex Sensor has promising potential for further development in various areas. Some future scope for this project includes:

Integration with Artificial Intelligence: The integration of artificial intelligence (AI) can enhance the capabilities of this system by enabling the robotic arm to learn and adapt to new tasks and environments. This could allow for more complex and control of machinery.

Improved Sensing Technology: Advancements in sensing technology could improve the accuracy and sensitivity of the flex sensors used in the glove. This could allow for even more precise control of the robotic arm.

Multi-Tasking: The ability to perform multiple tasks simultaneously could make the system more efficient and useful in industries such as manufacturing and logistics.

Extended Range: The development of long-range wireless communication technology could enable the system to control machinery over greater distances, expanding its potential use in remote and hazardous locations.

Increased Load Capacity: Increasing the load capacity of the robotic arm would make it more versatile and enable it to handle heavier machinery.

Applications in Medical Field: The system could be further developed for use in medical procedures, such as surgery, where precise and intuitive control of robotic instruments is crucial.

VIII. CONCLUSION

In conclusion, the use of a robotic arm controlled by a robotic glove has shown tremendous potential in a wide range of industries, including manufacturing, healthcare, and research. The ability to control a robotic arm through intuitive and natural hand movements allows for greater precision, dexterity, and flexibility in completing complex tasks, particularly in hazardous or difficult-to-reach environments. The technology continues to evolve and improve, with advances in sensors enabling even more sophisticated control and coordination between the glove and the arm. As a result, we can expect to see further integration of robotic glove- controlled arms in various applications, leading to increased efficiency and productivity.

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REFERENCES

- [1] R. Slyper and J. Hodgins, "Action Capture with Accel- erometers," Euro Graphics/ACMSIG GRAPHs Sym- posium on Computer Animation, 2008.
- [2] E. Foxl and L. Naimark, "Vis-Tracker: A Wearable Vi- sion-Inertial Self Tracker," IEEE Virtual Reality Confer- ence, 22-26 March 2003, Los Angeles.
- [3] M. Gross and D. James, "Eurographics/ACM SIGGRAPH Symposium on Computer Animation," Smart Objects Con- ference SOC '03, Grenoble, 2003.
- [4] L. Bio and S. Intille, "Activity Recognition from User-An- notated Acceleration Data," Pervasive Computing, Vol. 3001, 2004, pp. 1-17. doi: 10.1007/978-3-540- 24646-6 1
- [5] D. Fontaine, D. David and Y. Caritu, "Sourceless Human Body Motion Capture," Smart Objects Conference (SOC 2003), Grenoble, 2003.
- [6] J. Bernstein, H. R. Everett, L. Feng, and D. Wehe, "Mo- bile Robot Positioning Sensors & Techniques," Journal of Robotic Systems, Special Issue on Mobile Robots, Vol. 14, No. 4, pp. 231-249.
- [7] U. D. Meshram and R. Harkare, "FPGA Based Five Axis Robot Arm Controller," International Journal of Electron- ics Engineering, Vol. 2, No. 1, 2010, pp. 209-211.
- [8] <http://en.wikipedia.org/wiki/80c51>
- [9] C. Edwards and E. Smith, "Design of Simulink-Based 2- DOF Robot Arm Control Workstation," 31 October 2006. JSEA